

**AGENTIC ENGINEERING ASSESSMENT · SOFTWARE
EMPHASIS**

External signals × workforce retention

Asteria Consumer Products (fictional). How three retention targets are performing, and whether the economy adds useful context.

One command, runs offline

5 public signals, 2 providers

129 automated tests

Every number re-derived: 107 / 107

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The question, narrowed

Decision supported: **which retention target needs leadership attention, and does the economy explain it?**

TARGET	GOAL	HOW I DEFINED IT
New hires kept 6 months	$\geq 86\%$	Hires grouped by month from Jan 2021; "lost" = left before the 6-month anniversary. Only counted once hire + 6 months $\leq$ 31 Dec 2025; otherwise <i>not yet observable</i> , never "retained"
Senior hires kept 12 months	$\geq 90\%$	Same rule over 12 months, Senior Leader only ("Sr Mgmt" normalised to it; Managers only as a sensitivity check)
Regretted turnover, trailing 12 months	$\leq 7.5\%$	Regretted exits $\div$ mean of the 12 month-end active headcounts

The data is deliberately imperfect: nothing is dropped silently

23 quality rules, each with a severity that decides the row's fate. Counts reconciled with an independent profiling script.

ISSUE FOUND	ROWS	WHAT HAPPENED
Exact duplicates	7	Removed
Country "EL" / "ROM"	8	Mapped to GR / RO (EL is Eurostat's own code)
Blank country	9	Company-wide figures only
"Sr Mgmt" level	10	Normalised to Senior Leader (confirmed)
No hire date / left before hired	5 + 5	Excluded from all measures
Exit with no exit type	13	Counted as an exit, never as regretted
"End of Contract" on a Permanent	90	Kept, flagged
Voluntary exit, regretted unknown	2	Not regretted

2,407 → 2,400 → 2,390

rows read → people → measurable. 2,381 have a known country.

The trap: recent hires

Someone hired in October 2025 hasn't had 6 months yet. Counting them as "stayed" would inflate retention, so their group is **not yet observable**. That's why every hire-based headline covers **2021–2024**.

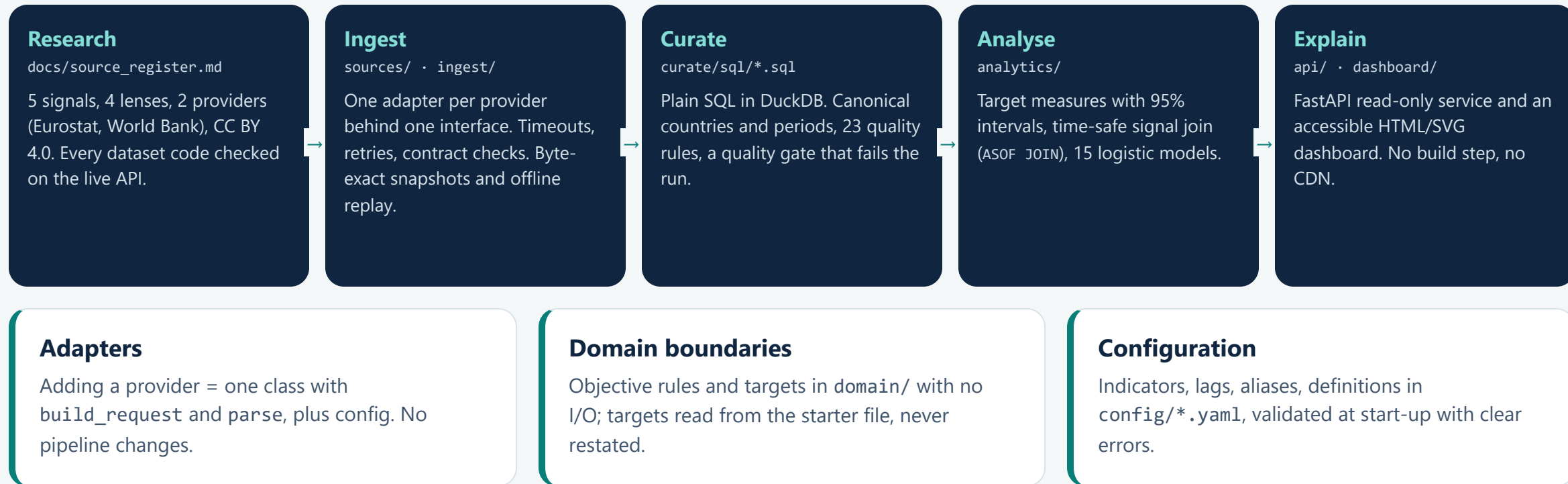
Also found

Nobody was hired before 2020, so headcount grows from 0 to ~1,600, which matters for reading turnover trends.

IN PLAIN WORDS We never quietly delete a messy row: each problem has a rule, and every rule's count is published.

One vertical slice, one command: `uv run asteria run`

Each stage is a separate module with its own tests; the core runs offline from committed snapshots in about 20 seconds.



Lineage: every value can be traced back to its source bytes

LAYER	WHERE	WHAT IT HOLDS	PROOF IT'S FAITHFUL
Raw	data/raw-or-fixtures/	Starter CSVs + API payloads, exactly as received	SHA-256 per file, checked by tests; <code>.gitattributes</code> stops line-ending edits
Source-shaped	DuckDB stg_*	Provider codes and period strings, untyped	Parsed only after a contract check (schema drift is an error)
Canonical	employees, external_observations	ISO countries, real period dates, quality flags	Lineage columns: source, dataset, period, loaded_at, snapshot sha
Analytical product	mart_*	Rates + intervals per country × unit × period; signals as known at the time; model results	Recomputed by separate code: 107 / 107 match

Deterministic

Reruns are **byte-identical**, including from a clean clone.

Idempotent

A second live ingest reports every source **unchanged**; nothing is rewritten.

Schema drift caught

Eurostat renamed `nace_r2` to `nace_r2_1`: the adapter treats a missing dimension as a contract error.

IN PLAIN WORDS For any number on the dashboard we can point to the exact downloaded file, and the exact rows, it came from.

No future information: a value is used only after it was published

SIGNAL	LAG	REAL RELEASES (SPOT CHECK)
Unemployment	62 d	30–33 days after month end
Job vacancies	90 d	75–79 days after quarter end
Inflation	25 d	17–24 days (January latest)
Economic sentiment	10 d	Before month end, except December: 5–8 Jan
GDP growth	200 d	Enters World Bank data on 1 July

Leak found and fixed: sentiment originally had a 0-day lag, so December values were used from 1 January, 5–8 days before publication (4,861 person-months). Fixed to 10 days; the conclusion didn't change.

What Greece "knew" on 1 June 2023

SIGNAL	VALUE	DESCRIBES	AGE
Sentiment	108.0	Apr 2023	32 d
Unemployment	11.4%	Mar 2023	62 d
Vacancies	0.9%	Q4 2022	152 d
GDP growth	8.7%	year 2021	517 d

Frequency integrity: an annual figure stays **labelled as its own year and age**, never shown as a monthly measurement.

Why it matters: the naive "same month" join produced the **strongest-looking result (p = 0.010)**. Leakage makes weak patterns look real.

IN PLAIN WORDS The analysis may only use economic news that had actually been published by that date, like reading yesterday's paper, never tomorrow's.

Reliability and engineering

Failure handling

- Transient errors (timeouts, 429, 5xx) retried with backoff; 4xx fail fast with the provider's message
- One failing source never stops or corrupts the others; its last good snapshot is kept
- A payload that breaks the contract never replaces good evidence
- API error contract: 404 / 422 (lists allowed values) / 503 "run asteria run" / 500 with request id, no internals

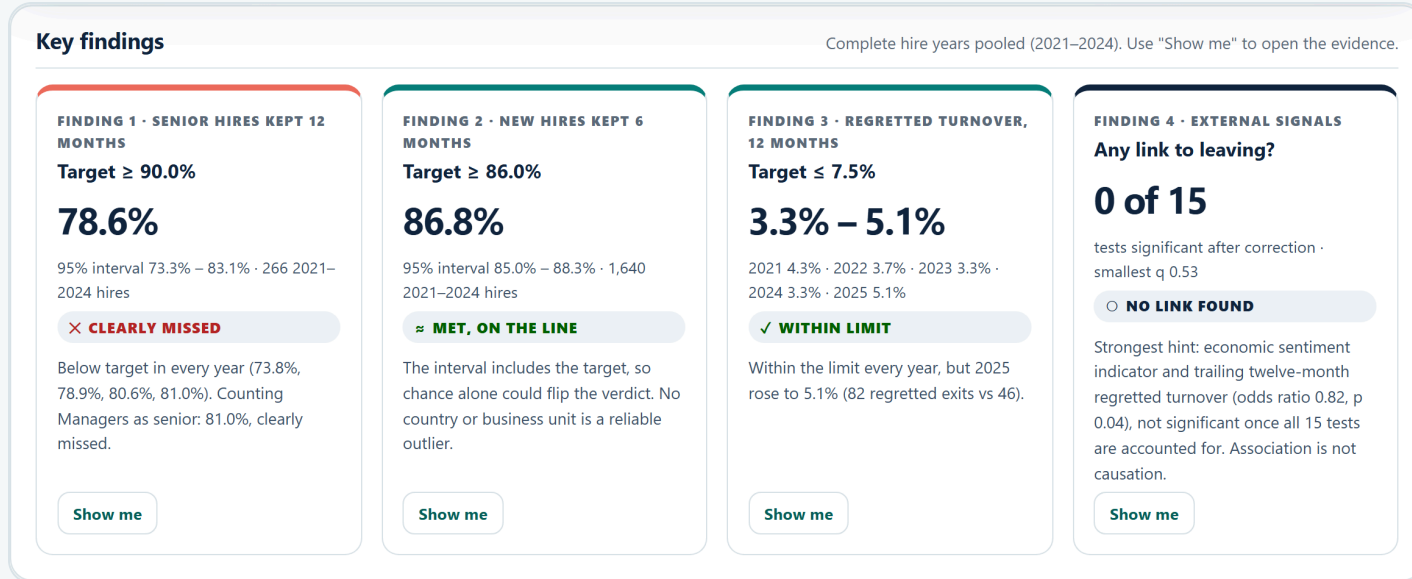
Observability

- Readable logs in a terminal, JSON when piped or scheduled
- Run manifest per ingest: attempts, checksums, coverage, warnings
- Quality report with every rule and its row count
- Dashboard "Trust" view: freshness, gaps, provider flags

TEST LAYER	WHAT IT PROVES
Unit + contract	Parsers, JSON-stat decoding, retries, raw store, config validation
SQL / metrics	Hand-computed cases: anniversary boundary, pending, headcount, Wilson interval, no-leak and period rules
Statistics	Known-answer tests: finds a planted effect; not fooled by country confounding
API + UI	Every endpoint and error; Playwright browser tests incl. empty/error states; axe accessibility audit in both themes

129 tests on Python 3.13 and 3.11 · strict mypy · ruff · tests validated by injecting bugs on purpose (each was caught)

The dashboard: findings first, then prove them



Explore

Objective, country, business unit, period, signal. Kept in the URL.

Understand

Trend with 95% band and target line, next to the signal as known each month.

Challenge

Country-quarter scatter sized by sample; naive vs within-country tests.

Trust

Freshness, coverage, exclusions, sources, method.

Accessible by design: keyboard-readable charts, a table for every chart, status as icon + text, validated colour-blind palette, graceful empty / pending / "too few people" / API-down states.

Reading the numbers, in plain words

Every figure on the next slides is one of these. Example used throughout: senior hires in 2024.

Rate	Of the 63 senior people hired in 2024, 51 were still here 12 months later: $51 \div 63 = \mathbf{81.0\%}$.
Pooled	Add the complete years together (people, not percentages): 209 stayed of 266 hired in 2021–2024 = 78.6% .
95% interval	How sure we are. With only 63 people the true rate could plausibly be anywhere from 69.6% to 88.8% ; more people give a narrower range.
"Clearly" vs "on the line"	Clearly : the whole range is on one side of the target. On the line : the range includes the target, so chance alone could flip the verdict.
Not yet observable	Too early to judge: hired less than 6 or 12 months before the data ends. Never counted as "stayed".
Too few people	Fewer than 30 people in a group: shown, but not judged. One person leaving would move the rate by more than 3 points.
Turnover	Regretted exits in a year $\div$ the mean of that year's 12 month-end headcounts: $82 \div 1,609.7 = \mathbf{5.1\%}$ in 2025.

Findings 1–3: how the targets are doing

Finding 1 · senior hires kept 12 months · target $\geq 90\%$

78.6%

95% interval 73.3–83.1% · 209 of 266 hires, 2021–2024

✗ **CLEARLY MISSED**

- Below target **every year**: 73.9, 78.9, 80.6, 81.0%
- Counting Managers as senior: 81.0%, still clearly missed
- Largest and most robust gap

Finding 2 · new hires kept 6 months · target $\geq 86\%$

86.8%

95% interval 85.0–88.3% · 1,423 of 1,640 hires

≈ **MET, ON THE LINE**

- Every year's interval includes 86%
- Countries 84.7–89.2%, units 85.1–88.8%: no reliable outlier
- Read "met" as "on the line"

Finding 3 · regretted turnover · limit $\leq 7.5\%$

3.3–5.1%

4.3 · 3.7 · 3.3 · 3.3 · **5.1%** (2021–2025)

✓ **WITHIN LIMIT, RISING**

- 2025: 82 regretted exits vs 46 in 2024
- Romania 6.7%, Ireland 6.2%, Bulgaria 6.1%: closest to the limit
- Caveat: young, growing workforce may explain part

How to read the verdicts: "**clearly**" means the whole 95% interval is on one side of the target (for senior hires even the top of the interval, 83.1%, is below 90%). "**On the line**" means the interval includes the target, so chance alone could flip it.

The economic question, in plain words

3 objectives = 3 different kinds of leaving

New hires leaving early, senior hires leaving early, and regretted voluntary exits. We test each separately, because the economy can push them in different directions.

5 signals × 3 objectives = 15 questions

Unemployment, job vacancies, inflation, economic sentiment and GDP growth, each tested against each objective.

ONE OF THE 15 QUESTIONS

"When unemployment is higher than usual in a country, are employees there more likely to have a regretted exit?"

Why we need leavers and stayers

Looking only at the people who left tells us what the economy looked like *when they left*, but not whether it made leaving **more likely**. For that we compare everyone at risk at the same moment: 65,184 person-months for turnover.

Three terms you'll hear

- **Odds ratio (OR):** 1.00 = no link; 0.82 = 18% lower odds of leaving per typical rise in the signal
- **p:** "could this one result be luck?" (small = unlikely luck)
- **q:** the same, after allowing for asking 15 questions. We need q below 0.05

Testing *any* exit is possible as an extra exploratory check, but it answers a broader question than the three objectives, so it isn't part of the findings.

THE KEY IDEA We ask whether changes in external conditions go together with changes in the likelihood of a specific type of leaving, comparing each country only with itself.

How each question becomes rows: person-months

For turnover, one row = **one employee in one month they were employed**.

Example: hired 2 May 2023, left 2 October 2023 (not regretted).

MONTH	EMPLOYED ON THE 1ST?	REGRETTED EXIT?	SENTIMENT KNOWN THEN
May 2023	no (hired on the 2nd): no row	–	–
June 2023	yes	0	108.0
July 2023	yes	0	107.1
August 2023	yes	0	109.0
September 2023	yes	0	110.1
October 2023	yes (left on the 2nd)	0 (not regretted)	110.7

Why rows per month

- **The economy changes monthly**, so each month's outcome is paired with that month's signal, as known at the time
- **Leavers vs stayers**: each month compares who left with everyone who could have left but stayed
- **Time at risk counts**: 40 months employed = 40 chances to leave; a new starter had 2

The hire targets are simpler

Everyone is followed over the same 6- or 12-month window, so it's **one row per hire** (1,808 new hires, 266 senior hires), matched to the economy in the hire month.

65,184 person-months · 217 regretted exits

Average chance of a regretted exit in any month: $217 \div 65,184 = 0.33\%$.

IN PLAIN WORDS In months when the economy looked like this, how often did people have a regretted exit, compared with everyone who could have left but stayed?

Inside the model: from rows to q

Worked on the strongest result: economic sentiment × regretted turnover.

1 · What goes in: one row per employee-month

INPUT	EXAMPLE
Left (regretted) this month?	0 or 1
Signal known then, in steps	$108.0 \div 5.78 = \mathbf{18.68}$
Months into the study	29 (June 2023)
Country	Greece
Constant	1

65,184 rows. Trend and country are **controls**, so time drift and country differences aren't credited to sentiment.

2 · COEFFICIENT

b = -0.20

The model's key output. Negative: higher sentiment, lower odds of a regretted exit

3 · ODDS RATIO

0.82

e^b . One step up (5.78 points) = 18% lower *odds*, not 18% lower probability

4 · 95% RANGE

0.68–0.99

Plausible values given the uncertainty

5 · P-VALUE

0.035

$z = b \div \text{std. error} = -2.11$. "If there were no link, how unusual is a result this strong?"

6 · Q-VALUE

0.53

Computed **after** all 15 models: adjusts p for testing 15 relationships

q = 0.53 is far above 0.05, so this is **not** counted as a link. The regression produces b, the odds ratio, the range and p; q comes from comparing all 15 results (Benjamini-Hochberg).

The flow: employee-month rows → regression → coefficient b → odds ratio + 95% range + p → adjust all 15 p-values → q. · **Simpler:** the model estimates the relationship; p says how surprising it is; q corrects for having asked 15 questions.

Finding 4: no economic signal lines up with leaving

0 of 15

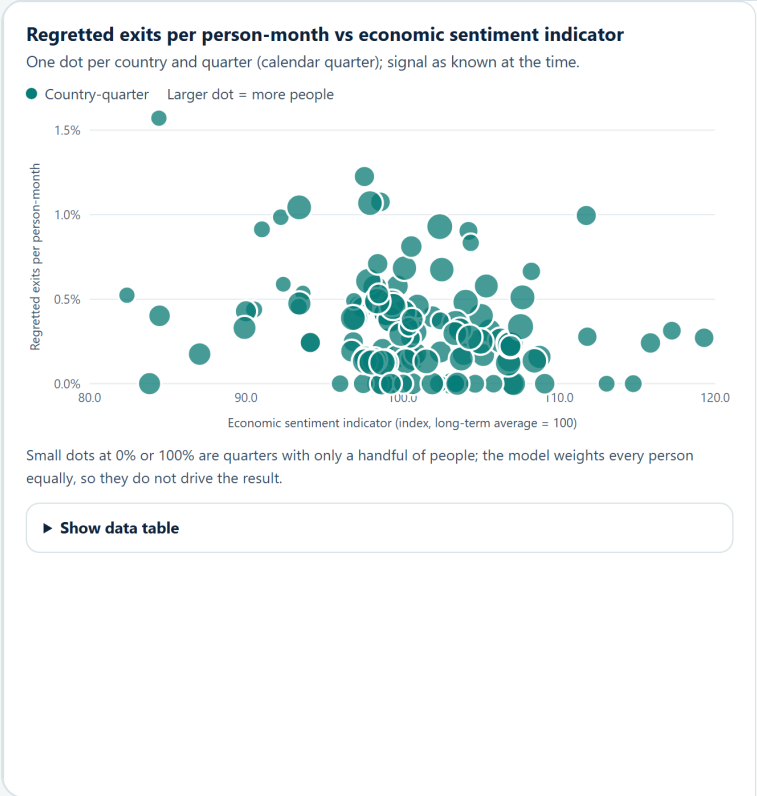
tests significant after correcting for running 15 tests · smallest q = 0.53 (needs < 0.05)

Method

- One row per person (or person-month), not averages
- Each country compared only with itself, plus a time trend
- Errors clustered by country-month; Benjamini-Hochberg correction

Why trust a "no"

- Same code finds a planted effect in synthetic data
- 8 alternative setups tested (timing, lags, controls, errors): none changed the answer
- Pooling countries flips unemployment's direction (0.97 → 1.11): that's the confounding the method removes



Is there an association?

Outcome: regretted exit in the month (per person-month at risk). Odds ratio per one typical within-country swing of the signal; 1.00 means no effect. "Within country" compares each country only with itself and removes a common time trend.

SIGNAL	NAIVE OR (95% CI)	WITHIN COUNTRY OR (95% CI)	Q	VERDICT
Economic sentiment indicator	0.87 (0.77–1.00)	0.82 (0.68–0.99)	0.53	Nominal only; not significant after correction
GDP growth	0.90 (0.79–1.02)	0.89 (0.77–1.02)	0.65	No evidence
HICP all-items	0.91 (0.80–1.04)	0.89 (0.75–1.05)	0.77	No evidence
Job vacancy rate	0.96 (0.88–1.05)	0.93 (0.78–1.12)	0.80	No evidence
Unemployment rate	0.97 (0.91–1.04)	1.11 (0.84–1.45)	0.80	No evidence

Only hint: higher business confidence, fewer regretted exits (odds ratio 0.82, p = 0.04). With 15 tests, a p this small is expected by chance, so q = 0.53. **No evidence of an effect is not evidence of no effect:** 57 senior leavers is a small sample.

How the conclusion was reached

ASTERIA RETENTION SIGNALS · HOW THE CONCLUSION WAS REACHED

From raw data to four findings

Left: the employee path (findings 1–3 use only this). Right: the external path, joined to employee outcomes to test finding 4.

1

INPUTS

Workforce file (synthetic)

2,407 rows: hires and exits across 6 countries, 2020–2025.
3 targets from the starter pack.

External data (public)

5 signals, 2 providers: unemployment, job vacancies, inflation, economic sentiment (Eurostat); GDP growth (World Bank).
2,456 values, monthly, quarterly and annual.

2

PREPARE

Clean and flag (23 quality rules)

- 7 duplicates removed → 2,400 people
- 10 invalid records excluded → 2,390 measurable
- 9 with no country: company-wide figures only

Time-safe join ("what was known then")

- A value is used only after its publication date
- Lags: 62 / 90 / 25 / 10 / 200 days, checked against real release dates
- Annual values keep their own year and age

3

MEASURE AND TEST

Measure the three targets

- Group hires by month; lost = left before the 6- or 12-month anniversary
- Unfinished groups are not yet observable, never counted as retained
- Turnover = regretted exits ÷ mean of the 12 month-end headcounts
- 95% interval on every rate; fewer than 30 people = not judged

Test: does the economy line up with leaving?

- 15 models (3 targets × 5 signals), one row per person
- Each country compared only with itself, plus a time trend
- Correction for running 15 tests (Benjamini-Hochberg q)

4

FINDINGS (POOLED 2021–2024 HIRES; TURNOVER PER CALENDAR YEAR)

FINDING 1 · SENIOR HIRES KEPT 12 MONTHS

Target ≥ 90%

78.6%

95% interval 73.3–83.1% · 266 hires

✗ CLEARLY MISSED

Every year below target (73.9 → 81.0%). Still missed if Managers count as senior (81.0%).

FINDING 2 · NEW HIRES KEPT 6 MONTHS

Target ≥ 86%

86.8%

95% interval 85.0–88.3% · 1,640 hires

≈ MET, ON THE LINE

Every year's interval includes 86%. No country or business unit is a reliable outlier.

FINDING 3 · REGRETTED TURNOVER, 12 MONTHS

Limit ≤ 7.5%

3.3–5.1%

4.3 · 3.7 · 3.3 · 3.3 · 5.1% (2021–2025)

✓ WITHIN LIMIT, RISING

2025 up (82 exits vs 46). RO, IE, BG closest to the limit. Growing, young workforce may explain part.

FINDING 4 · EXTERNAL SIGNALS

Any link to leaving?

0 of 15

significant after correction · smallest q 0.53

○ NO ASSOCIATION

Only hint: higher sentiment, fewer regretted exits (p 0.04), not significant after correction. Association ≠ causation.

107 / 107

every number recomputed by separate code

124 tests

metrics, time rules, API, dashboard, accessibility

Lags spot-checked

against official release dates; one leak found and fixed

Byte-identical

rebuild from a clean clone

Definitions confirmed by the clarification answers: senior = Senior Leader (Sr Mgmt normalised to it); hires from 2021 with complete windows; turnover denominator = mean of 12 month-end headcounts. Workforce data is synthetic.

Tradeoffs and limitations

Limits of the evidence

- **Synthetic, hired from 2020 on:** a turnover trend may reflect a longer-serving workforce, not the economy
- **6 countries, ~5 years:** large effects are detectable; small ones may go unnoticed
- **Time since hire not controlled:** new starters leave far more (2.4% a month in the first 6 months vs 0.5% after 2 years)
- **Revised economic data:** most signals are today's revised figures; inflation was checked and never revised
- **Italy's vacancies:** measured differently, so not fully comparable

Deliberate choices

- **DuckDB + FastAPI + HTML:** free, offline, easy to review; can move to Databricks + Power BI
- **Conservative lags:** never use data before it was public; the cost is slightly staler data
- **One lag per series:** consistent method; Ireland's earlier publishing not exploited
- **Logistic regression, not ML:** the question is "is there a link?", not "who will leave?"
- **Full rebuild:** 2,400 rows rebuild in seconds; incremental loading isn't worth the complexity

IN ONE SENTENCE We prioritised fairness, interpretability, reproducibility and no data leakage; the small, synthetic dataset limits how far the results generalise.

How I used the AI agent, and stayed accountable

Decisions that stayed mine

- Software emphasis; repository shape (changed the agent's layout to the brief's)
- **Created CLAUDE.md**: every rule from the brief, loaded each session to keep the agent in scope; extended when it slipped (e.g. Python 3.11 syntax)
- Cut clarification questions 10 → 3; analyse blind, not from the data generator
- Indicator set (from my own search, fact-checked by the agent)
- Methodology: when I found I hadn't been consulted, every option was put back to me and re-run
- Lag fixes, dashboard changes, pooling rule
- **API refactor**: split the single-file API into routers, repository, schemas and services

Agent errors caught by verification

- Defect list from reading code: 3 overstatements, 1 missed; fixed by profiling
- Two licence URLs recalled from memory were wrong
- 0-day sentiment lag leaked December data; found by the release-date spot check
- Interval formula crashed on tiny slices; found by its own test
- Accessibility contrast failures; found by the axe audit

Verification, not trust: tests, an independent recomputation of every number (107 / 107), live checks against provider APIs and release dates, screenshots reviewed. All logged in AI_USAGE.md, including my own corrections.

Next steps

#	NEXT STEP	WHAT IT MEANS
1	Add time since hire	Account for how long someone has worked there: new employees are much more likely to leave
2	Act on senior retention	Investigate why senior hires leave: onboarding, role fit
3	Release-calendar ingest	Record the actual publication date of every economic value, instead of one fixed delay
4	Point-in-time vintages	Keep the value that was available at the time, not today's revised history
5	Robustness checks	Try other statistical methods and see whether the conclusions hold
6	Productionise	Move from DuckDB / FastAPI / HTML to ADF + Databricks + Power BI

Try it

```
git clone ../asteria-retention-signals
uv sync · uv run asteria run · uv run asteria serve
```

Read more

[docs/findings.md](#) · [docs/analysis_explained.md](#) · [docs/source_register.md](#) · [docs/architecture.md](#) · [AI_USAGE.md](#)